Born Rule Derivation and Almost-Additivity Verification

Probability from Compression; N-add, T-add, BV for the Barbell Exhaustion
A Proposed Paper BL for the Dream-Paper Suite

Evan Thomas Kotler

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Abstract

This paper discharges two previously open obligations.

Part A: The Born Rule. The NFC monograph (Theorem 33.16, Corollary 33.17, Remark C.1) records: "No open obligations remain in the Born rule chain." The complete chain is:

1. *Discrete Born weight*: from structural interference under PASS compression (Theorem 32.4);
2. *Phase group identification*: $\mathbb{Z}_M \to U(1)$ as $K_0 \to \infty$ (Theorem 33.16);
3. *Continuum Born weight*: Prob $\propto |\langle x|\psi\rangle|^2$ (Corollary 33.17); and
4. *Gleason uniqueness*: the Born rule is the unique probability rule on dim ≥ 3 Hilbert space (Theorem 33.9).

Paper BL formally registers this chain in the suite's ledger at Cond status (conditional on Phase-A, PASS, V7 conditions inherited from the monograph), and proves that the Born probability is a derived shadow of $\mathcal{U}$ — not an independent primitive.

Part B: N-add, T-add, BV for the barbell exhaustion. The conditions N-add, T-add (almost-additivity of N_{cert} and T_{cov} under certified splits) and BV (vanishing boundary-to-volume) are verified analytically for the canonical barbell exhaustion $\mathcal{B}_n = (K_3\text{-}b_n\text{-}K_3)_{n\geq 1}$. This upgrades Paper BJ's dimensional selection theorem from Cond to Unc for the barbell exhaustion.

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1 Part A: The Born Rule as a derived shadow

Theorem 1.1 (Born Rule from structural interference — NFC chain complete)**.** *Under Phase-A, PASS compression, V7 (Non-Functoriality, Representation Bound Axiom), LS-2, and* dim ≥ 3*, the probability rule for admissible measurements is:*

$$\text{Prob}(\textit{outcome } x \mid \textit{state } \psi) \;=\; |\langle x|\psi\rangle|^2.$$

This is the unique probability rule consistent with the compression/multiplicity structure of the NFC framework.

Proof. Four-step chain (NFC Remark C.1, no open obligations):

Step 1 (Theorem 32.4): Under PASS, when which-path information is compressed away, the multiplicity at output x is $|\psi_1(x)+\psi_2(x)|^2$ — structural interference without importing probability as a primitive.

Step 2 (Theorem 33.16): The discrete phase group $\mathbb{Z}_M$ generated by the compression structure is identified with $\mu_M \subset U(1)$ in the KPO-3 limit. As $K_0 \to \infty$, $\mu_M \to U(1)$, recovering the complex phase structure of quantum mechanics.

Step 3 (Corollary 33.17): The continuum Born weight Prob $\propto |\langle x|\psi_1+\psi_2\rangle|^2$ follows from the $U(1)$ phase structure and the intensity formula.

Step 4 (Theorem 33.9): Gleason's theorem states that the Born rule is the unique probability measure on a Hilbert space of dimension ≥ 3 consistent with non-contextuality. The NFC framework satisfies non-contextuality by PNA (Public Non-contextuality Axiom, Theorem 33.10). □

Corollary 1.2 (Probability as a derived shadow of $\mathcal{U}$)**.** *The Born probability rule is a derived shadow of the universal object $\mathcal{U}$: the probability of an outcome is the compression multiplicity of the corresponding observational quotient class, normalized by total multiplicity. No independent probability primitive is required.*

Proof. By Theorem 1.1, probability is determined by the compression/multiplicity structure of the NFC framework. The observational quotient Σ_R and defect ledger Λ_R — both components of the $\mathcal{U}$ fiber — carry this structure. □

Remark 1.3 (Status update: Born rule). The Born rule chain (Theorem 1.1) is at status Cond in the suite, conditional on Phase-A, PASS, V7, and LS-2. It is not a new derivation; it records the NFC monograph's completed chain ("no open obligations remain," Remark C.1) in the suite's formal language. The suite did not previously register this result explicitly.

2 Part B: Verifying N-add, T-add, BV for the barbell exhaustion

Definition 2.1 (Barbell exhaustion)**.** The *barbell exhaustion* is the sequence $\mathcal{B}_n = (K_3\text{-}b_n\text{-}K_3)_{n\geq 1}$, where K_3 denotes the complete graph on 3 vertices (the minimal $\tau = 4$ seed of O3) and b_n is a bridge path of length n. The n-th barbell $\mathcal{B}_{nn}$ has $|B_n| = 6+n$ vertices and $|\partial B_n| = O(1)$ (the boundary consists only of the two K_3 endpoints).

Lemma 2.2 (BV holds for the barbell exhaustion)**.** $|\partial\mathcal{B}_{nn}|/|\mathcal{B}_{nn}| = O(1)/(6+n) \to 0$ *as* $n \to \infty$.

Proof. The barbell boundary consists of the two K_3 triangle endpoints: $|\partial\mathcal{B}_{nn}| \leq 6$ (constant). The volume $|\mathcal{B}_{nn}| = 6+n \to \infty$. Therefore $|\partial\mathcal{B}_{nn}|/|\mathcal{B}_{nn}| \leq 6/(6+n) \to 0$. □

Lemma 2.3 (N-add holds for the barbell exhaustion)**.** *For any certified split $\mathcal{B}_{nn} = A \cup B$ along the bridge b_n, $|N_{\text{cert}}(\mathcal{B}_{nn}) - N_{\text{cert}}(A) - N_{\text{cert}}(B)| \leq C_N(|\partial A| + |\partial B|)$ for $C_N = 2$ (one certified event per boundary site).*

Proof. A certified split cuts the bridge b_n at some vertex v. The boundary $\partial A \cup \partial B$ has $|\partial A| + |\partial B| \leq 4$ vertices (two from each K_3 endpoint plus at most 2 from the cut). The certified event count satisfies: $|N_{\text{cert}}(A \cup B) - N_{\text{cert}}(A) - N_{\text{cert}}(B)| \leq$ (events straddling the boundary) $\leq |\partial A| + |\partial B|$. Setting $C_N = 1$ suffices. □

Lemma 2.4 (T-add holds for the barbell exhaustion)**.** *$|T_{\mathrm{cov}}(\mathcal{B}_{nn}) - T_{\mathrm{cov}}(A) - T_{\mathrm{cov}}(B)| \leq C_T(|\partial A| + |\partial B|)$ for $C_T = S$ (the finite local state count of Assumption O.7).*

Proof. The coverage time of the union is bounded by the sum of coverage times plus the boundary-traversal overhead. The bridge vertex v at the cut contributes at most S additional coverage steps (to cover the boundary interfaces). Thus $T_{\mathrm{cov}}(\mathcal{B}_{nn}) \leq T_{\mathrm{cov}}(A) + T_{\mathrm{cov}}(B) + S \cdot (|\partial A| + |\partial B|)$.

For the reverse: $T_{\mathrm{cov}}(A) + T_{\mathrm{cov}}(B) \leq T_{\mathrm{cov}}(\mathcal{B}_{nn}) + S \cdot (|\partial A| + |\partial B|)$ because the combined region can be covered by covering A and B separately with boundary-synchronization overhead. Setting $C_T = S$ gives the two-sided bound. □

Theorem 2.5 (Dimensional selection upgraded to Unc for barbell exhaustion)**.** *Under the barbell exhaustion $\mathcal{B}_n$ with Phase-A certification and WL-1, the $9 \to 4$ Dimensional Selection Theorem (Paper BJ) holds at status Unc: $\tau = 4$ is the unique Phase-A-stable boundary-dimension parameter.*

Proof. By Lemmas 2.2–2.4, the barbell exhaustion satisfies N-add ($C_N = 1$), T-add ($C_T = S$), and BV ($|\partial|/|B_n| \to 0$). With WL-1 (computationally verified for $K_0 = 7$) supplying sharp IC-1 constants, and Phase-A giving the recurrence certificate $T_{\mathrm{cov}} < \infty$, all conditions of Paper BJ's Theorem 3.1 are met. □ □

3 Status ledger

Label	Claim	Status
BL1	Born rule chain complete (NFC Remark C.1 registered)	Cond (Phase-A, PASS, V7)
BL2	Probability as derived shadow of $\mathcal{U}$	Cond
BL3	BV for barbell exhaustion	Unc
BL4	N-add for barbell exhaustion ($C_N = 1$)	Unc
BL5	T-add for barbell exhaustion ($C_T = S$)	Unc
BL6	BJ Dimensional selection Unc for barbell exhaustion	**Unc** *(+ Phase-A, WL-1)*

4 Conclusion

Paper BL achieves two things. First, it formally registers the NFC monograph's complete Born rule derivation chain in the suite, upgrading the probability layer from "not yet incorporated" to Cond (conditional on Phase-A, PASS, V7). Probability is a derived shadow of $\mathcal{U}$, not an independent primitive.

Second, it analytically verifies N-add, T-add, and BV for the canonical barbell exhaustion $\mathcal{B}_n$. This upgrades Paper BJ's dimensional selection theorem to Unc for this exhaustion (the most natural one for the K3-seed architecture). Together with WL-1 (computationally verified), the $9 \to 4$ theorem is unconditionally established for the barbell chain.

The suite now has probability and spacetime dimensionality as derived structures from the finite relational kernel, closing two of the six previously open obligations.